

IoT-Based Heart Attack Pattern Detection Using Deep Learning

Chapters 1 to 3: Introduction, Literature Review, and Methodology

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Chapter 1: Introduction

1.1 Background

Heart disease is one of the biggest causes of death in the world. The World Health Organization says that heart and blood vessel diseases cause about one third of all deaths, and most of these are because of heart attacks and strokes [1]. A heart attack, which doctors call a myocardial infarction (MI), happens when the blood flow to a part of the heart gets blocked. When this happens, that part of the heart starts to get damaged. The faster the problem is found and treated, the more of the heart can be saved. So finding heart problems early is very important and can save many lives.

The main tool that doctors use to check the heart is the electrocardiogram, or ECG. An ECG records the small electrical signals that the heart makes. When something is wrong with the heart, the shape of the ECG signal changes in certain ways. A trained doctor can look at these changes and tell if a person is having a heart attack. The problem is that ECG machines in hospitals are big and expensive, and you need a trained person to use them. Because of this, most people cannot check their heart all the time. They usually only go to the hospital after they already feel very sick.

Two new technologies can help solve this problem. The first one is the Internet of Things (IoT). IoT means small and cheap devices that can collect data and send it over the internet. A small sensor called the AD8232, together with a low-cost chip called the ESP32, can record an ECG signal and send it wirelessly for only a few dollars [2], [3]. The second technology is deep learning, which is a type of artificial intelligence. Deep learning models, and especially ones called convolutional neural networks (CNNs), have been able to read ECG signals almost as well as expert doctors [4], [5].

This research brings these two ideas together. The goal is to build a cheap device that records the ECG signal with a single sensor, and a deep learning model that looks at the signal and finds heart attack patterns. The result is shown on a simple web page in real time. In this way, more people, even in faraway villages, could check their heart without going to a big hospital.

1.2 Problem Statement

Even though ECG is a very good way to find heart problems, there are still some big problems that stop normal people from getting heart checks all the time. Hospital ECG machines cost a lot of money, they cannot be moved easily, and they need a trained person to use them. So most people have no way to watch their heart all the time, and they only get checked when things are already bad. This is even worse for people who live far from hospitals or who do not have much money [1].

At the same time, most of the strong deep learning research on ECG was made and tested using full 12-lead hospital recordings that were taken in clean conditions [5], [6]. Only a small number of studies check if a single-lead signal, like the one a cheap IoT sensor gives, has enough information to find a heart attack in a reliable way. And even fewer studies put the model inside a full system that works in real time. So there is a gap between what works in the lab with clean hospital data and what can really be done with a cheap device in everyday life.

So the main problem this research looks at is this: there is no cheap, real-time system that joins a low-cost single-lead IoT ECG device with a deep learning model that can find heart attack patterns and quickly warn the user. To solve this, we need to find out if cheap hardware can record a good enough signal, if a deep learning model can find heart attacks from one lead, and if the whole system is fast enough to be useful.

1.3 Research Aim

The aim of this research is to design, build, and test a low-cost IoT system that uses deep learning to detect heart attack patterns from a single-lead ECG signal, and to show the heart risk to the user in real time on a web page.

1.4 Research Objectives

To reach this aim, the research works through these steps, one task at a time:

- 1 Review past research on deep-learning ECG analysis, single-lead heart attack detection, and IoT-based ECG monitoring, and find the research gap this study fills.
- 2 Prepare a single-lead (Lead I) Normal versus heart attack dataset from the PTB-XL database, using patient-disjoint splits.
- 3 Design a 1D convolutional neural network (1D CNN) for single-lead ECG classification.
- 4 Train the 1D CNN on the training set.
- 5 Choose the classification threshold using the validation set.
- 6 Test the trained model on the unseen test set.
- 7 Build a low-cost single-lead ECG device using an ESP32 microcontroller and an AD8232 sensor.
- 8 Develop a Flask web application that runs the model and shows the heart-risk result.
- 9 Measure the end-to-end speed (latency) of the whole system.

1.5 Research Questions

This research tries to answer these questions:

- 1 **RQ1:** Can a cheap ESP32 and AD8232 device record a single-lead ECG signal that is good enough to use?
- 2 **RQ2:** Can a deep learning model find heart attack patterns from a single-lead ECG signal?
- 3 **RQ3:** Is the server (cloud) prediction fast enough for real-time heart monitoring?

1.6 Scope of the Study

This research is about finding heart attack patterns from single-lead ECG data using deep learning, and showing the result through an IoT device and a web page. To keep the work clear and possible to finish, the scope is kept simple. The model is mainly trained and tested on the PTB-XL dataset [6]. The task is a simple two-class problem: telling a heart attack signal apart from a normal signal, because this is the most useful and most realistic target for an undergraduate project. The

hardware is built as a simple proof-of-concept device, not as a real medical machine.

This study does not try to find every kind of heart problem, and it does not claim to be as good as a hospital. It does not do big tests on many patients. Any signals taken from the device are only used to test the prototype, and they come from the researcher or from people who agree to take part, under proper ethical rules. The system is meant to be an early-warning and research tool, not a replacement for a real doctor.

1.7 Significance of the Study

This research is useful in two ways: in real life and in academic study. In real life, it helps build a cheap way to watch the heart, which could bring heart monitoring to poor and faraway places where normal machines are too expensive. Because it supports remote monitoring, it fits well with the growing use of online health care, and it could warn people early so they go to a doctor in time.

In academic study, the work adds to the small area of cheap, single-lead ECG analysis with deep learning, and it helps connect hospital-level research with technology that normal people can afford [7], [8]. It also shows a full, working system, from the sensor, to sending the data, to running the model, to showing the result, instead of only looking at the model alone. Finally, the device can be a base for future wearable heart-monitoring products that other people can improve later.

1.8 Thesis Structure

The rest of this thesis is set out as follows:

- **Chapter 1, Introduction**, gives the background, the problem, the aim, the objectives, the research questions, the scope, and why the study is important.
- **Chapter 2, Literature Review**, looks at past research, the basic theory of ECG and deep learning, a review of the good and weak points of past work, and the research gap this study fills.
- **Chapter 3, Methodology**, explains the research design, the system, how the data is collected and cleaned, how the model is built, the tools used, how the system is tested, and the ethical rules.
- **Chapter 4, Implementation (later)**, will explain how the hardware, the model, and the web app are built.
- **Chapter 5, Results and Evaluation (later)**, will show and explain how well the system works.
- **Chapter 6, Conclusion and Future Work (later)**, will sum up the findings, talk about the limits, and suggest future ideas.

Chapter 2: Literature Review

2.1 Introduction

This chapter looks at the past research that is connected to this project. First it explains the basic theory, which means what an ECG signal is, how a heart attack changes the signal, and how deep learning works for this kind of data. Then it looks at past studies in three groups: deep learning for ECG, single-lead and low-cost ECG, and IoT ECG systems. After that, it points out the good and weak points of these studies, and finally it explains the research gap that this project fills.

2.2 Theoretical Foundations

The ECG and heart attacks. An ECG is a recording of the small electrical signals made by the heart. One heartbeat makes a wave shape with three main parts: the P wave, the QRS complex, and the T wave. The size and shape of these parts tell us if the heart is healthy. When a person has a heart attack, these parts change in known ways, such as a higher or lower ST segment, a flipped T wave, or a deep Q wave. Because these changes show up in different places, hospitals use a 12-lead ECG that looks at the heart from many sides. Some leads are better than others for seeing certain heart attacks [7]. An important question for cheap systems is whether one lead still has enough information to find a heart attack.

Deep learning for signals. In the past, people used hand-made features, like measuring the size and timing of the waves, and then gave these to simple machine learning models. This works, but it needs an expert to choose the features. Deep learning is different because the model learns the useful features by itself from the raw signal. The convolutional neural network (CNN) is very good for ECG. A 1D CNN slides small filters along the signal and learns local shapes, like the shape of the QRS complex, step by step. Some studies also use LSTM models or mix CNN and LSTM to catch longer patterns in time [9]. These models are the base of modern ECG research. A large systematic review by Hong et al. [10], which looked at almost two hundred studies, confirms that CNN-based deep learning is now the most common and most successful way to work with ECG data.

2.3 Related Studies

Deep learning for ECG. One of the first big works in this area was by Rajpurkar et al. [4]. They trained a deep CNN to find heart rhythm problems from a single-lead ECG, and the model worked almost as well as expert heart doctors. This showed that deep learning can read ECGs well without hand-made features, as long as there is enough data. This work was later improved and published by Hannun et al. [5] in *Nature Medicine*, where a deep CNN sorted twelve types of heart rhythm and did better than the average doctor on the same test. These two studies are seen as very important examples of computer ECG reading. More recently, Ribeiro et al. [11] trained a deep neural network on more than two million ECG exams and showed that a computer can correctly find several heart problems from the 12-lead ECG, which proved that these methods also work on very large, real-world data.

Big, well-labelled public datasets made this progress possible. The PTB-XL dataset by Wagner et al. [6] has 21,837 hospital 12-lead ECG records from 18,885 patients. Each record is ten seconds long and was labelled by up to two heart doctors, and it has a special group for heart attacks and many normal records too. Because of this, PTB-XL is now a common standard that lets researchers compare their methods in a fair way. Building on this dataset, Strodthoff et al. [12] tested several deep learning models on PTB-XL and gave standard benchmark results, which make it easier to compare new methods like the one in this project.

Single-lead and low-cost ECG. Some early work already showed that deep CNNs can find heart attacks from ECG signals. Acharya et al. [13] used a CNN on a single ECG lead (Lead II) and reached about 93 percent accuracy for heart attack detection, and Baloglu et al. [14] built a deep CNN that found heart attacks from multi-lead ECG signals. Newer research then checks if the good results from 12-lead recordings also work with only one lead, which is what cheap devices give. A 2025 study tested several deep learning models on single leads to tell apart normal, abnormal, old heart attack, and new heart attack signals, and it reached F1-scores as high as 98 percent on the best leads, with very fast prediction times [7]. This study also showed that the choice of lead really matters. Other work looked at heart attack detection from Lead I, which is the lead that most wrist and two-electrode devices can get, and showed that it is possible, but usually with lower sensitivity than using many leads [8]. So these studies show that single-lead heart attack detection can work, but it is harder than the full 12-lead case and depends a lot on signal quality and lead choice.

IoT ECG systems. Another group of research is about the hardware that records and sends the ECG outside the hospital. A broad review of ECG monitoring systems by Serhani et al. [15] explains the common parts and the main problems of these systems, from the sensor to the cloud. Many projects have shown ECG recording using the cheap AD8232 sensor with low-cost chips like the ESP32 and ESP8266, sending the data over WiFi to the cloud [2], [3]. These works prove that the recording part is cheap and possible, but many of them stop at sending and showing the data and do not add any automatic detection. A smaller number of studies add intelligence to the system. For example, one embedded system found atrial fibrillation in real time using an AD8232 sensor and a CNN-based model on a Raspberry Pi [9]. Other deep learning IoT systems watch health data from wearable sensors to find health problems early [10], [16]. These are the closest works to this project, but they mostly look at heart rhythm or general health, not heart attacks.

2.4 Critical Analysis

When we look at all this research together, we can see three different groups that this project wants to join. The first group, deep learning for ECG, has made very accurate models, but almost all of them were trained and tested on clean hospital 12-lead data [4], [5], [6]. Their high scores may not stay the same on the noisy single-lead signals that a cheap device gives.

The second group, single-lead ECG, has started to fix this and has shown that single-lead heart attack detection is possible. But a lot of this work still uses clean single leads taken from hospital datasets like PTB-XL, not real signals from cheap sensors [7], [8]. So there is still a gap between the data used for training and the data the system would really see when used in the field.

The third group, IoT ECG hardware, has clearly shown that cheap recording and sending is possible [2], [3], but many of these projects stop at showing the signal and leave the reading to a doctor. When they do add intelligence, it is most often for heart rhythm problems [9], [16], not for heart attacks, which are a different and very time-critical problem. Also, not many studies report how fast the whole system is, even though speed is very important for an early-warning tool. So the research gives strong but separate pieces: good models, cheap hardware, and new single-lead methods that have not yet been joined into one heart-attack-focused, real-time, low-cost system.

2.5 Research Gap

Putting all of this together, the research gap that this project fills is the lack of one full, cheap, single-lead system that uses deep learning to find heart attack patterns in real time. Past work has shown, on its own, that deep learning can read ECGs very well, that single-lead heart attack detection is possible, and that IoT hardware can record and send ECG signals cheaply. But no well-known study has joined these three things into one full system that is (i) made just for heart attacks, (ii) built on cheap single-lead hardware, and (iii) tested for speed as well as accuracy. This project fills that gap by building and testing exactly this kind of system.

2.6 Novel Contribution of the Study

What is new in this research is that it brings together three things that earlier studies did only on their own, and joins them into one complete system. Earlier work has shown that deep learning can read ECGs well, that single-lead heart attack detection is possible, and that cheap IoT hardware can record and send ECG signals. But no well-known study has joined all three into one working system.

This research is new because that single system:

- is made just for heart attacks, not for heart rhythm problems like most earlier smart IoT ECG systems;
- uses a single lead (Lead I), the lead that cheap two-electrode devices can record, instead of a full 12-lead hospital recording; and
- is tested for both accuracy and speed (end-to-end latency), not accuracy alone.

In short, the novelty is a complete, low-cost system that collects a single-lead ECG, detects heart-attack patterns using deep learning, shows the risk in real time, and is judged on both how well and how fast it works.

2.7 Summary

This chapter explained the basic theory of ECG and deep learning, looked at the three main groups of past research, talked about their good and weak points, and showed the research gap. The next chapter explains the method used to fill this gap.

Chapter 3: Methodology

3.1 Introduction

This chapter explains how the system is designed, built, and tested. It talks about the research design and why it was chosen, the system parts, how the data is collected and cleaned, how the deep learning model is built, the tools that are used, how the system is tested, and the ethical rules that are followed.

3.2 Research Design

This study uses a quantitative experimental design. It is quantitative because the main claims, about how accurate the model is and how fast the system is, are tested using numbers from experiments. It is experimental because the system is built and then tested in a careful way against data it has not seen and under measured conditions. This lets the research questions be answered with real evidence and not just opinion.

The main idea (hypothesis) is that a deep learning model can find heart attack patterns from single-lead ECG signals with high accuracy, and that the IoT system around it can give results fast enough for real-time use. The design follows the normal machine learning way of working: the model is trained on a training set, tuned on a validation set, and tested on a test set that it has not seen. Together with this, the hardware and the web app are built and their full speed is measured.

3.3 System Design

The system is made of five steps that move the data from the body to a final heart-risk result shown to the user:

AD8232 ECG Sensor → ESP32 (WiFi) → Flask Server → 1D CNN Model → Web Dashboard (Risk Output)

Layer	Part	What it does
Sensing	AD8232 + 3 electrodes	Records the single-lead ECG signal from the body
Edge / IoT	ESP32 chip	Turns the signal into data and sends it over WiFi
Server	Flask app (Python)	Takes the data, cleans it, and gives it to the model
Intelligence	1D CNN (TensorFlow / Keras)	Tells if each ECG part is Normal or a heart attack
Display	Web dashboard	Shows the live ECG, the risk status, and alerts

The work is done step by step in an iterative way. The model and the hardware are built at the same time and joined slowly. This is safer, because the software can still move forward even if the hardware is late or the signal is noisy. This is possible because the system has a simulation mode that can replay saved dataset signals instead of the live sensor. Each part is tested on its own before the full system is put together.

3.4 Data Collection

Main dataset: PTB-XL. The main data for training and testing the model is the PTB-XL dataset [6], a large, free, hospital-labelled 12-lead ECG database from PhysioNet. It has 21,837 ten-second records from 18,885 patients, labelled by up to two heart doctors, with a special group for heart attacks and many normal records. PTB-XL is chosen because it is big enough to train deep models, it is free for research, it is a common standard so results can be compared with other work, and it has the heart attack and normal groups that this study needs.

To make the data look like the single-lead signal from the AD8232 sensor, only one lead (Lead I) is taken from each 12-lead record for training. This makes the model learn from the same kind of view that the real device gives, so the test is more realistic.

Extra data: device signals. As well as the public dataset, a small amount of ECG data is taken from the hardware prototype to test and show the system. These recordings are taken from the researcher and from people who agree, under the ethical rules in Section 3.9. This extra data is used to check that the hardware gives a good signal (this answers RQ1) and to show the full system, not to train the model.

3.5 Data Preprocessing and Model Building

Raw ECG signals must be cleaned before the model can use them. The same steps are used on both the training data and the live data, so the model sees the same kind of input in training and in real use:

- **Filtering:** a band-pass filter removes slow drift and high-frequency noise and keeps the useful part of the ECG.
- **Normalisation:** each signal is scaled so the model is not confused by different signal sizes from different devices.
- **Segmentation:** long signals are cut into fixed-length windows that fit the model input.
- **Class balancing:** because there are far more normal records than heart attack records, class weights are used so the model does not just learn to say normal every time.

The data is split into about 70 percent training, 15 percent validation, and 15 percent test. The split is done by patient, so records from the same patient are not in two sets at once. This stops data leakage, which would make the results look better than they really are.

The model is a 1D convolutional neural network (1D CNN) built with TensorFlow and Keras. It has a few convolution blocks (each with a 1D convolution, batch normalisation, a ReLU activation, and pooling), then some dense layers, and a final layer that gives the chance of a heart attack. The 1D CNN is chosen because it learns the shapes of the ECG wave by itself, it has worked well in past ECG research [4], [5], [7], and it is light enough to run fast on the server.

Training uses binary cross-entropy loss and the Adam optimiser, with dropout and early stopping to stop overfitting. The settings, like the number of filters and the learning rate, are tuned using the validation set. The final model is saved and loaded once when the server starts, so it adds very

little delay.

3.6 Tools and Technologies

The tools used in this project are listed below. They are mostly free, open-source, and low-cost, which fits the goal of making a cheap system.

Area	Tool	Why it is used
Hardware	ESP32, AD8232, electrodes	Cheap single-lead ECG recording and sending
Firmware	Arduino IDE (C/C++)	Reading the signal and sending it over WiFi
Dataset	PTB-XL (PhysioNet)	Training and testing data
ML library	Python, TensorFlow / Keras	Building and training the 1D CNN
Signal tools	NumPy, SciPy, WFDB, NeuroKit2	Loading and cleaning the ECG signals
Web app	Flask, Chart.js / Plotly	Server, dashboard, and live charts
Communication	HTTP REST / MQTT	Sending data from ESP32 to the server

3.7 Evaluation Metrics

The system is tested in two ways that match the research questions. Accuracy of detection (RQ1, RQ2) is measured on the test set using accuracy, precision, recall (also called sensitivity), and the F1-score, plus the confusion matrix and the ROC-AUC. Precision and recall are shown together with accuracy because, when one class is much bigger, accuracy alone can be misleading. In a health system, recall is very important, because missing a real heart attack is more dangerous than a false alarm.

Speed (RQ3) is measured as the time from when the ESP32 sends the data to when the risk result shows on the web page. A delay of a few seconds or less is taken as good enough for real-time monitoring.

Metric	What it means	Why it matters
Accuracy	How many predictions are correct	General correctness
Precision	Correct alarms out of all alarms	Fewer false alarms
Recall / Sensitivity	Real heart attacks that were found	Catching true cases
F1-score	Balance of precision and recall	Fair score when classes are uneven
ROC-AUC	How well it separates the classes	Score not tied to one threshold
Latency	Time from signal to result	Shows if it works in real time

3.8 Research Outcomes

The outcome of this research is a complete low-cost heart-attack pattern detection system. An ECG device was built using an ESP32 and AD8232 sensor to collect real single-lead ECG data from a consenting participant or heart patient. The ESP32 sends the ECG data to the server through Wi-Fi. The deep-learning model then analyses the ECG signal for heart-attack patterns, and the web application displays the ECG signal and risk result in real time.

The study used 14,538 ECG recordings and an ensemble of five 1D CNN models. The system achieved a ROC-AUC of 0.9076, detected 503 out of 550 heart-attack cases, achieved 91.5% recall, and produced a prediction in about 0.35 seconds.

3.9 Ethical Considerations

Even though this project is mostly technical, it uses human heart data and is about health, so some ethical points must be followed. First, the system is clearly a research prototype and not a real medical device. This is stated wherever results are shown, and the system is not used in place of a real doctor.

Second, the PTB-XL dataset is fully anonymous and is free for research, so it has no privacy problem. Third, if any ECG signals are recorded from the researcher or other people using the device, they take part by choice and they give informed consent first. They are told why the recording is done and that they can stop at any time, and any data is kept anonymous and stored safely, following the rules of the university. Finally, the limits of the system, such as the chance of missing a heart attack, are shown clearly so the technology is not over-claimed.

3.10 Summary

This chapter explained the quantitative experimental method of the study. It described the five-step system, the use of the PTB-XL dataset together with some device signals, how the data is cleaned and the model is built, the tools used, the test metrics linked to the research questions, and the ethical rules. The next chapters will show how this method is built and what results it gives.

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